

#	Area	Reference business logic	jsonprocessor behavior	Finding
1	Syntax	N/A	Missing comma after "AVM_YEAR_BUILT": payload.get("AVM_YEAR_BUILT")	● BUG — file does not parse. jsonprocessor L108–109.
2	Payload access	Normal fields are read normally	Uses payload.geT(...) for condo fee, monthly tax, and rate type	● BUG — dict has no geT method. After fixing #1, these lines will throw AttributeError. jsonprocessor L126–128.
3	Utility import path	/opt/ml/processing/input/util	/opt/ml/perocessing/input/util	● BUG — typo perocessing can prevent util import depending on deployment. jsonprocessor L14–24.
4	Credit-score validity	Scores <300 or >998 → missing	Uses .between(300,998)	✓ Same rule. Reference: borrower_processor L252–263. JSON L170–187.
5	Credit-score binning	np.digitize(..., right=False)+1	np.searchsorted(..., side="right")+1	✓ Equivalent at the cutpoints. Reference L280–295. JSON L205–207.
6	Borrower score priority	First available bureau according to configured priority	First available bureau according to configured priority	✓ Same, assuming JSON config contains exactly the same

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				priority. JSON L190–203.
7	Application which_score	Sort ordered which_score and retain the highest- priority score source represented among borrowers	Uses statistical mode of which_score	 BUSINESS LOGIC DIFFERENCE. Two borrowers using CV and one using TV would return TV in original logic but CV in JSON. Original behavior is in borrower application aggregation around L370–376. JSON uses mode at L163–168 and L209–216.
8	Application CRD_SCORE	Numeric fields initially aggregate by max()	borrower_scores.max()	 Same for CRD_SCORE. JSON L155–167.
9	Application CRD_SCORE_b in	Application numeric aggregation uses max; subsequently rounded	JSON uses ceil(max(bin))	 Effectively same because bins are already integers, but ceil() is unnecessary.
10	Credit-score statistics	Produces CRD_SCORE_min, max, avg, median, wavg, and related bin statistics	Not generated	 MISSING BUSINESS LOGIC. CRD_SCORE_avg and CRD_SCORE_max are part of downstream final data.
11	Weighted score	Weight by BASE_SALARY,	Not calculated	 MISSING.

#	Area	Reference business logic	jsonprocessor behavior	Finding
		falling back to TOTAL_INCOME		
1 2	Top-two TV scores	Generates TV_Credit_Score_S MI3.0_B1 and B2 from highest two borrower TV scores	Not generated	● MISSING. Original borrower aggregation includes this logic.
1 3	NUM_OF_BOR	Count borrower rows for each application	len(b)	✓ Same if JSON BORROWERS contains exactly one row per actual borrower. JSON L145–168.
1 4	NUM_OF_BOR _bin	1→1, 2→2, >2→3 using [0,1,2,∞]	Not generated	● MISSING MODEL FEATURE.
1 5	TU score cleanup	If TU score absent, TU months/open-trades/public-derog are cleared	Not implemented	● MISSING borrower preprocessing. Reference borrower_processor L264–269.
1 6	TU/EQ fallback	Missing TU months/open-trade/public-derog filled from EQ equivalents	Not implemented	● MISSING. Reference L271–276.
1 7	Collection	Satisfied OR unsatisfied collection == "Y" → "Y"	Same basic rule, but JSON uppercases values first	● JSON is slightly more permissive ("y" also counts). JSON L218–229.

#	Area	Reference business logic	jsonprocessor behavior	Finding
1 8	Judgment	Satisfied OR unsatisfied judgment == "Y" → "Y"	Same basic rule, case-insensitive	 Slight behavioral difference only for lowercase input. JSON L231–242.
1 9	Units missing-value handling	Numeric conversion; 3/4→"3-4"; missing restored to NaN	Converts entire column to string before recoding	 BUG: missing values can become literal "nan" instead of actual NaN. JSON L244–251. Original utility explicitly restores missing values to NaN.
2 0	Units values >4	Numeric 5.0 becomes string "5"	5.0 stays "5.0"	 DIFFERENCE that can create unseen categorical levels.
2 1	DSR difference	abs(TDSR-GDSR), rounded 3 decimals, same bins	Calls the same derive_dsr_features()	 Same. JSON L254–256. Reference utility L260–279.
2 2	Recent/testing DSR source	Testing/recent processing replaces TDSR/GDSR with PRD_COMP_TDSR/GDSR	Uses payload TDSR/GDSR directly	 BUSINESS LOGIC DIFFERENCE for production/recent scoring.
2 3	Property-age imputation	Missing age = BOOKYEAR - AVM_YEAR_BUILT	Uses year from Date_Credit_Engine – AVM year	 DIFFERENCE. JSON L259–301.
2 4	New construction	PRK_CONS_TYPE=="N" → age 1	Same, additionally strips/uppercases input	 Same intended result. JSON L292–300.

#	Area	Reference business logic	jsonprocessor behavior	Finding
2 5	Existing property value validity	Existing value retained only when >1	Same	 Same. JSON L305–324.
2 6	Property- value fallback	If current invalid, fallback to purchase price then declared value	JSON also requires fallback purchase/declared values themselves to be >1	 DIFFERENCE. Original logic only applies >1 validity check to existing PRK_PROPERTY_VALU E; JSON applies it to all three candidates.
2 7	Condo fee	Missing/non- numeric condo fee → 0 for everyone	Missing fee → 0 only when PRK_TENURE != "C"; condo records remain missing	 BUSINESS LOGIC DIFFERENCE. JSON L329–341.
2 8	Province	Policy preprocessing receives actual PROVINCE from source data	JSON derives province from first postal-code character	 NEW/DIFFERENT RULE. Potential mismatch if postal code is invalid or application province disagrees with postal code. JSON L344–375.
2 9	Province from postal code	N/A in original policy processing	Does not map Yukon/NWT/Nunavut explicitly ; returns OTHERS	 Territories ultimately become Territories, but original province itself is lost.
3 0	Economic province mapping	Province → configured region; otherwise Territories	Same general behavior	 Assuming config.yaml mapping matches PROVINCE_REGION_M AP. JSON L377–395.

#	Area	Reference business logic	jsonprocessor behavior	Finding
3 1	Toronto FSA loading	Intended source logic extracts the FSA column for records identified as Toronto	JSON simply loads every value from the first CSV column	● LIKELY DIFFERENCE/BUG , depending on FSA_T26.csv layout. JSON L79–90.
3 2	Toronto override	Ontario + Toronto FSA → Toronto; other Ontario → Rest_of_Ontario	Same once self.toronto_fsa is correct	✓ Same intended rule. JSON L383–394.
3 3	Toronto file failure	Original flow expects Toronto mapping data	Any exception silently returns empty set	● RISK : every Ontario application silently becomes Rest_of_Ontario. JSON L82–90.
3 4	Economics historical/future cutoff	History through 202211; forecast beginning 202212	Same	✓ Same. JSON L421–471.
3 5	HPI shifts	future HPI/current HPI	Same	✓ Same. JSON L463–469.
3 6	UE +4	Future UE – current UE	Same	✓ Same. JSON L471.
3 7	Inference economic date	Testing and recent populations force region HPI/UE values to the 202212 reference values	JSON merges economics using configured effect_sid	● Potentially major difference . It only matches if config["economics"]["effect_sid"] == 202212. JSON L474–488.
3 8	Contract rate for recent/test	Preprocess overwrites contract rate with the	JSON takes value from payload	● DIFFERENCE .

#	Area	Reference business logic	jsonprocessor behavior	Finding
		configured fixed contract rate		
3 9	Contract-rate fallback	Missing contract rate → qualifying rate	Uses payload.get("PRD_CONTRACT_RATE", payload.get("PRD_QUALIFYING_RATE"))	 BUG for explicit nulls. If JSON contains "PRD_CONTRACT_RATE": null, Python returns None; it does not fall back to qualifying rate. JSON L119–122.
4 0	Insurance-in-force fallback	Missing IIF → purchase price × ACTLTV/100 → base-loan amount	Not implemented	 MISSING. JSON merely reads POD_INSURANCE_IN_FORCE.
4 1	ACTLTV_bin	Generated using configured cutpoints	Never generated	 MISSING MODEL FEATURE. Reference utility L286–305.
4 2	Feature clipping	Training/testing/recent values clipped to defined model ranges	No clipping	 MISSING. This can create out-of-training-range inference values.
4 3	Required model fields	Rows lacking bankruptcy flag, score bin, market code, property age, port, borrower count are excluded	JSON does not validate these fields	 MISSING inference validation.
4 4	Rural indicator	Derived from postal-code second character, with NB city override	Not generated	 MISSING feature.

#	Area	Reference business logic	jsonprocessor behavior	Finding
4 5	Loan-purpose bin	Refinance "R"→1, otherwise 0	Not generated	● MISSING preprocessing rule.
4 6	HPI_change meaning	Main LGD processing uses separately configured hpi_change; backtest LGD uses HPI_+4	JSON simply sets HPI_change = HPI_+4	● SEMANTIC DIFFERENCE. JSON L397–401.
4 7	EFFECT_DAY_SID	Source policy retains/repairs actual effect day	JSON fabricates the first day of configured month: effect_sid*100+1	● DIFFERENCE. JSON L92–101.
4 8	HPI log transform	Shared utility defines a consistent $\log_1 p(\max(\text{HPI}_+4, 0))$ transformation	JSON returns raw HPI_+4	● Verify where the model pipeline applies this. If not applied later, inference will differ from model preprocessing.
4 9	Product/population filters	Training/testing apply government-guarantee/product/refinance-balance filters	JSON performs none	● For a pure scoring endpoint this may be intentionally omitted, but it is not business-logic equivalent .
5 0	Status/category/target	Original policy flow derives funded/declined status, event category, and target	Not present	● Usually unnecessary at inference, but again not equivalent to full preprocessing.

The most critical parity problems

For **model-scoring parity**, I would fix these first:

1. **Syntax/runtime blockers** — missing comma and all three payload.get() calls.
jsonprocessor
2. **which_score aggregation** — use source priority, **not mode**. jsonprocessor
3. Generate missing model features: **CRD_SCORE_avg**, **CRD_SCORE_max**, **NUM_OF_BOR_bin**, **ACTLTV_bin**, **TV B1/B2**, and any borrower variables expected by the model.
4. Apply the **same recent/test DSR behavior** (PRD_COMP_TDSR/GDSR if this JSON processor represents production/recent inference).
5. Apply **feature clipping** with the same training bounds.
6. Match the **202212 economic-scenario freeze** used by test/recent preprocessing rather than simply joining on arbitrary configured effect_sid.
7. Match the **fixed contract-rate** behavior and correct null fallback.
8. Add **POD_INSURANCE_IN_FORCE** imputation.
9. Fix **condo-fee** behavior to match preprocessing.
10. Fix **units missing handling** so NaN does not become the categorical string "nan".
11. Decide whether property age should use **BOOKYEAR/effect year** rather than Date_Credit_Engine.
12. Fix/verify the **Toronto FSA loader**.

Component	Parity
Credit score validation	✓ Good
Score bin calculation	✓ Good

Component	Parity
Per-borrower score priority	✓ Good
Application borrower aggregation	✗ Different
DSR difference/bin	✓ Good
Property age	⚠ Different reference year
Property value	⚠ Slightly different
Condo fee	✗ Different
Units recode	✗ Missing-value bug
Economic feature construction	✓ Mostly same
Economic scenario used for scoring	✗ Potentially different
Contract rate	✗ Different
ACTLTV bin	✗ Missing
NUM_OF_BOR bin	✗ Missing
Feature clipping	✗ Missing
Insurance-in-force imputation	✗ Missing
Rural indicator	✗ Missing
Credit-score application statistics	✗ Missing
File executable as uploaded	✗ No